



Faculty of Information Technology

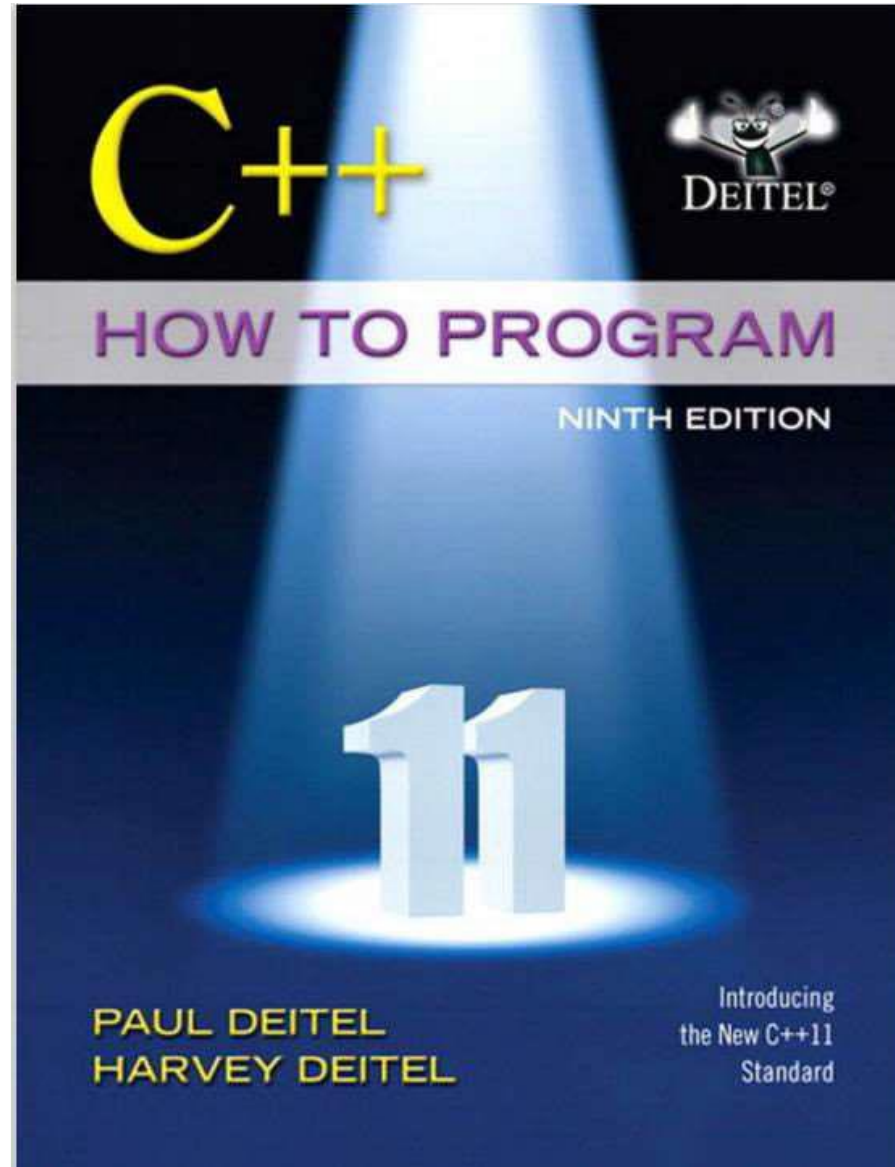
Fall 2026-2027

Introduction to Computers Programming CS.102

By Dr. Doha Taha

Assessment Schedule

Schedule	Assessment Method		Assessment Timing (Week Number)	Marks/ Scores	Percentage of total course Marks
	Semester Work	Exam (1)	6	30	20 %
		Exam (2)	11	30	20 %
	Final Practical Exam		12	30	20 %
	Final Written Exam		14	60	40 %





Introduction to Computers and C++

Chapter 1 of C++ How to Program, 11/e

INTRODUCTION

- C++ is one of today's most popular software development languages.
- You'll write instructions commanding computers to perform those kinds of tasks
- C++17, C++20, and C++23, the most recent of which is C++23, are the latest versions standardized through the International Organization for Standardization ISO and the International Electrotechnical Commission IEC

Computer Program

- **A *computer program*** is a sequence of instructions written to perform a specified task with a computer.
- **The program** has an executable form (i.e., after you write the program to perform a specific task and then save it, **.exe** file is obtained for running)
- The **.exe** program:
 - It has human human-readable form called the source code
 - this source code is written by computer programmers.
 - This source code was written using a programming language.
 - An **interpreter or compiler** is used to convert the source code into **.exe** form.

Programming Languages

- It is an artificial language designed to communicate instructions to a computer.
- It has a text form (i.e., it contains words, numbers, and punctuation).
- Have three groups: machine language, assembly language, and high-level language.

Programming Languages

- *machine language:*

- program code has 0's and 1's only.
- difficult to read and write.

- *assembly language:*

- Its level higher than machine language.
- program code has mnemonic English codes. (ADD, SUM, SUB etc)
- needs **assembler** to translate assembly code into machine code.

- *high-level language (HHL)*

- Its level higher than machine or assembly language.
- program code has human language.
- examples of HHLs are C, C++, VB, Java ... etc.
- needs **compiler** or **interpreter** to translate HHL code into machine code.

Evolution of Computer Programming Languages

Early beginnings (1940s–1950s)

- 1st Generation: Machine Language. In the earliest days of computing, programmers had to write instructions directly in binary code (strings of 0s and 1s).
- 2nd Generation: Assembly Language (1949). Assembly language replaced binary code with mnemonics (short, symbolic codes) that were more readable for humans.
- FORTRAN (1957). Developed by IBM, FORmula TRANslation was one of the first high-level languages with a functional compiler.

Evolution of Computer Programming Languages

- FORTRAN (1957). Developed by IBM, FORmula TRANslation was one of the first high-level languages with a functional compiler.
- COBOL (1959). Created for business applications, the Common Business-Oriented Language focused on data processing and was designed to be highly readable using English-like syntax.
- LISP (1959). Developed for artificial intelligence research, LISP (List Processor) introduced key concepts like recursion and symbolic computation.

The era of structured programming (1960s–1970s)

- BASIC (1964). Beginners' All-purpose Symbolic Instruction Code was designed to be simple and accessible for students and non-experts.
- Pascal (1970). Developed as a teaching tool.
- C (1972). Created at Bell Labs, C combined low-level memory access with high-level features.

The era of structured programming (1960s–1970s)

- object-oriented and internet boom (1980s–1990s)
- C++ (1983). Developed as an extension of C, C++ added object-oriented features.
- Python (1991), Java, JavaScript, and PHP (1995)

Modern era (2000s–present)

- C# (2000). Microsoft's answer to Java, C# is an object-oriented language for building applications on the .NET framework.
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- Swift (2014). Developed by Apple to replace Objective-C for iOS and macOS development, it emphasizes safety and modern features.

Advantages of high-level language

- **Readability:** Programs are easy to read.
- **Maintainability:** Programs are easy to maintain.
- **Portability:** Programs are easy to port across different computer platforms
- **Reusability:** Programs are saved into a library file and invoked them in next programming project simply by including the library file.

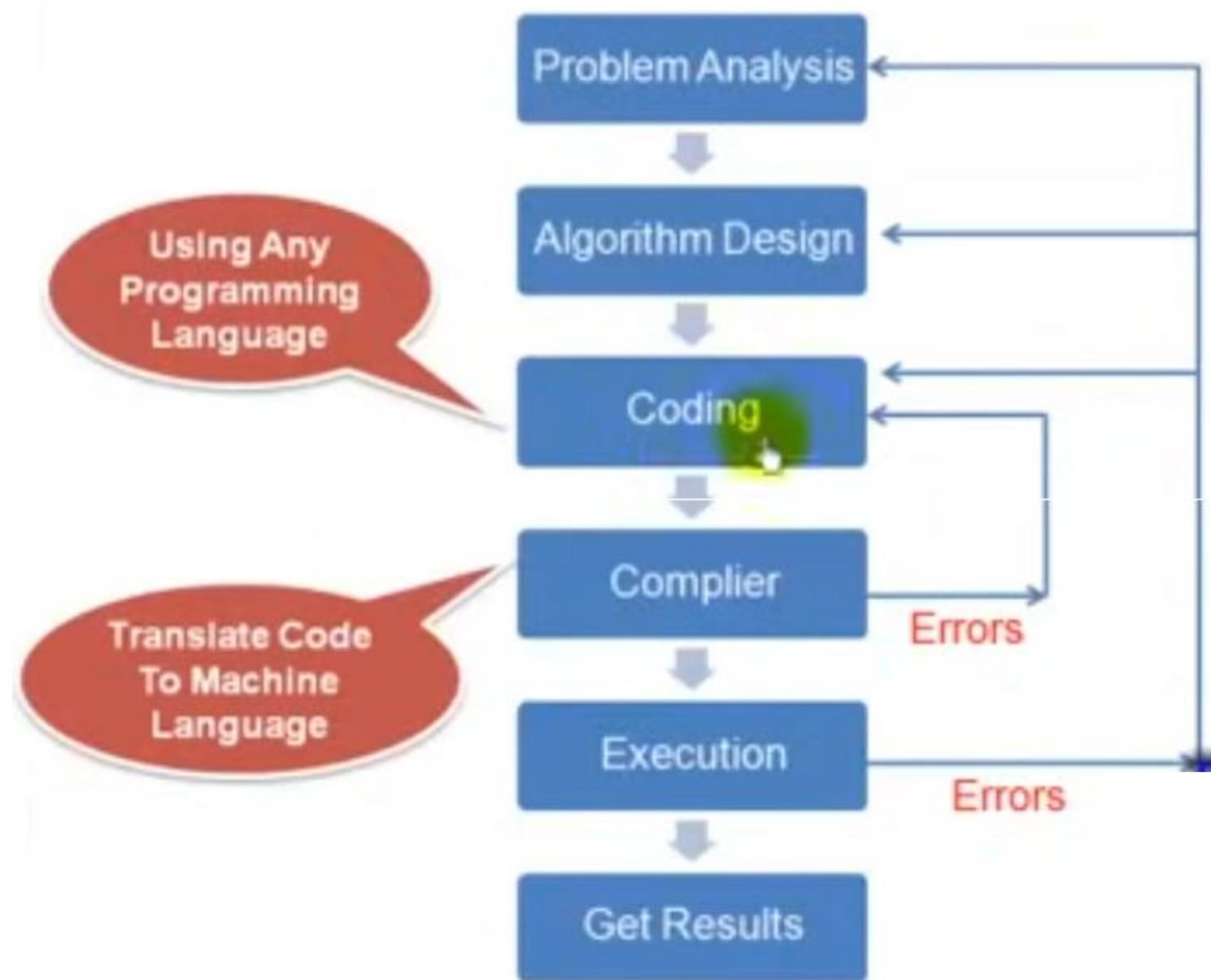
What is programming and the program development life cycle?

- **Programming** is a process of problem solving.
- **Step 1: Analyze the problem**
 - Outline the problem and its requirements.
 - Design steps(algorihm) to solve the problem.
- **Step 2:implement the algorihm**
 - Implement the algorithm in code.
 - Verify that the algorithm works.
- **Step 3: Maintenance**
 - Use and modify the program if the program domain changes

□ **Algorithms**

- Step-by-step problem–solving process.

The Problem Analysis–Coding–Execution Cycle





Think

Then



Write Code

Basic Definitions

Programming language:

- a set of rules, symbols, and special words used to write computer programs.

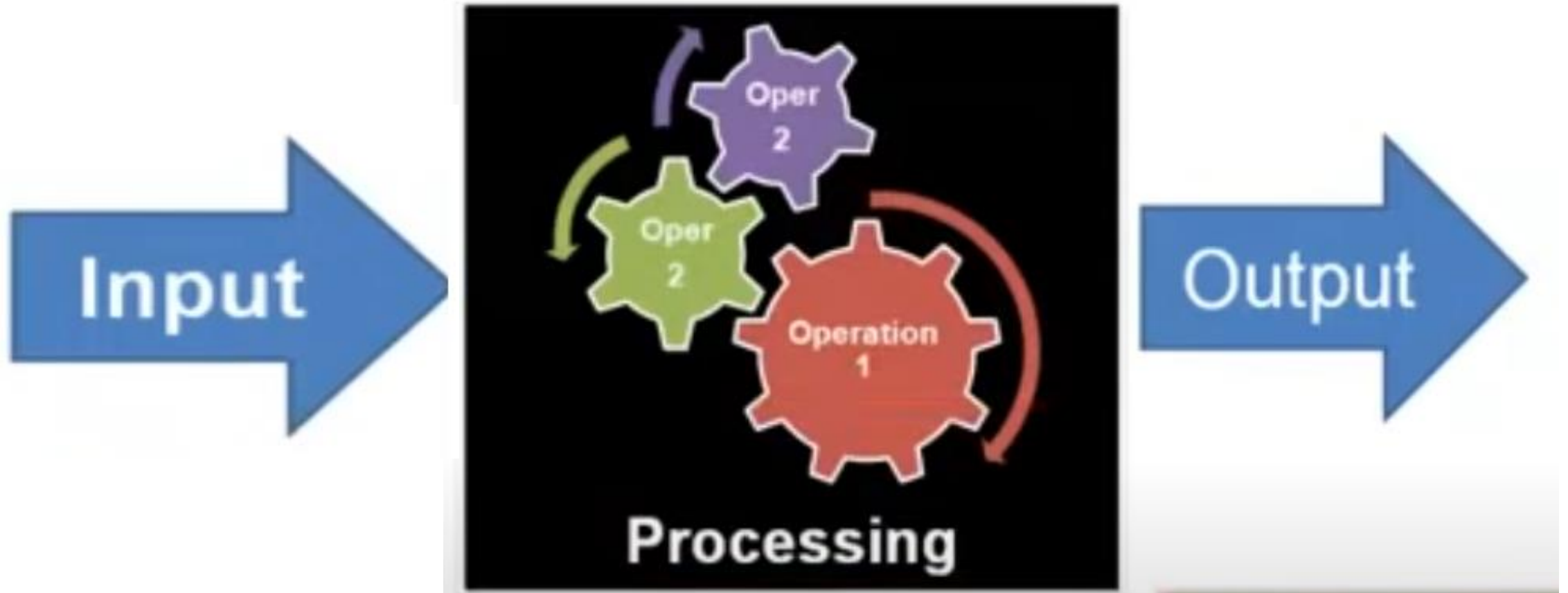
Computer program

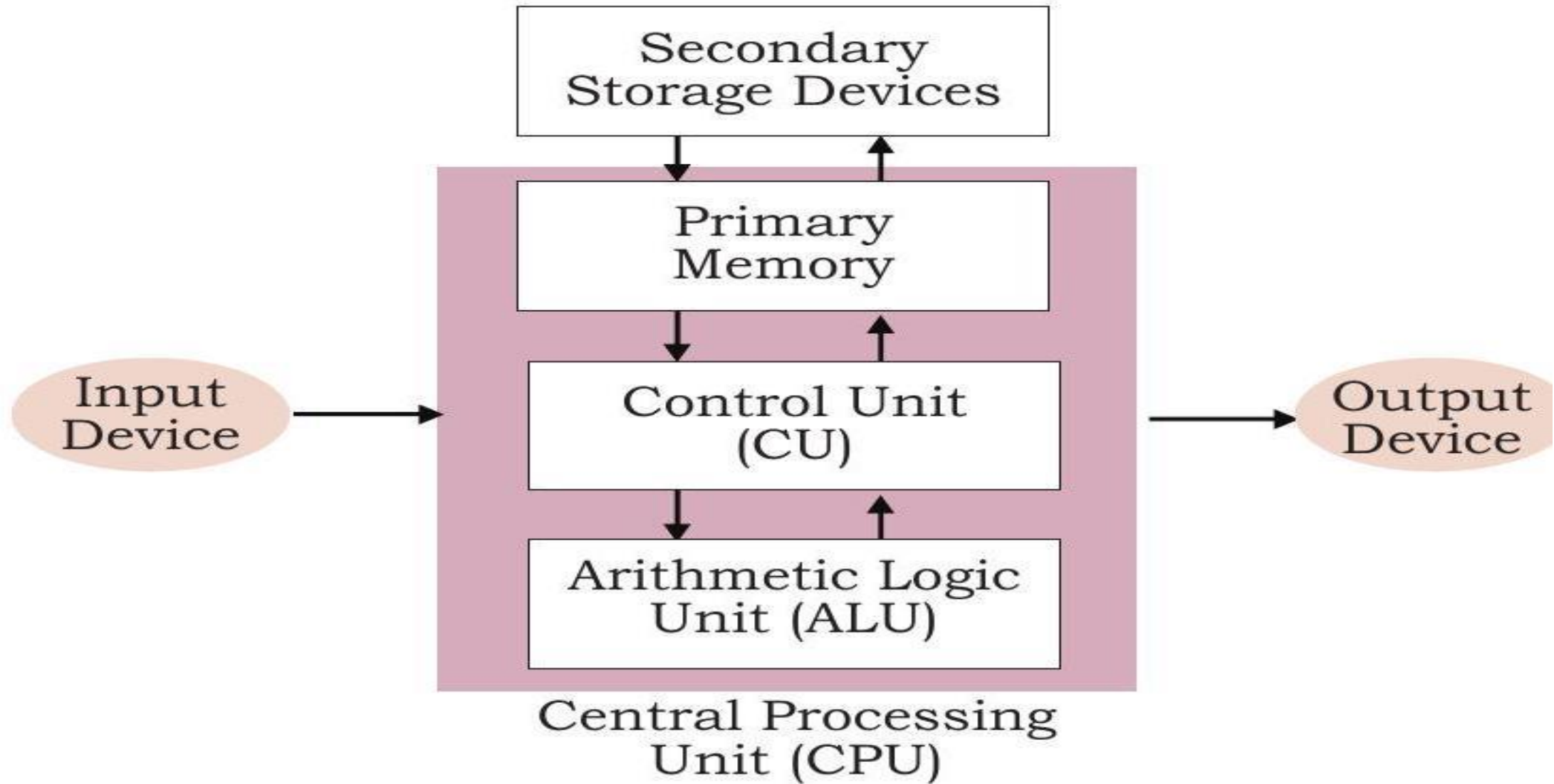
- Sequence of statements whose objective is to accomplish a task.

Syntax:

- rules that specify which statements (instructions) are **legal**

Computer system





Hardware and Software

- Computers process data under the control of sequences of instructions called **computer programs**.
- These programs guide the computer through ordered actions specified by people called computer **programmers**.
- The programs that run on a computer are referred to as **software**.

Logical unit	Description
Input unit	This “receiving” section obtains information (data and computer programs) from input devices and places it at the disposal of the other units for processing. Most user input is entered into computers through keyboards, touch screens and mouse devices. Other forms of input include receiving voice commands, scanning images and barcodes, reading from secondary storage devices (such as hard drives, DVD drives, Blu-ray Disc™ drives and USB flash drives—also called “thumb drives” or “memory sticks”), receiving video from a webcam and having your computer receive information from the Internet (such as when you stream videos from YouTube® or download e-books from Amazon). Newer forms of input include position data from a GPS device, and motion and orientation information from an <i>accelerometer</i> (a device that responds to up/down, left/right and forward/backward acceleration) in a smartphone or game controller (such as Microsoft® Kinect® for Xbox®, Wii™ Remote and Sony® PlayStation® Move).

Fig. 1.2 | Logical units of a computer. (Part 1 of 5.)

Logical unit	Description
Output unit	This “shipping” section takes information the computer has processed and places it on various output devices to make it available for use outside the computer. Most information that’s output from computers today is displayed on screens (including touch screens), printed on paper (“going green” discourages this), played as audio or video on PCs and media players (such as Apple’s iPods) and giant screens in sports stadiums, transmitted over the Internet or used to control other devices, such as robots and “intelligent” appliances. Information is also commonly output to secondary storage devices, such as hard drives, DVD drives and USB flash drives. Popular recent forms of output are smartphone and game controller vibration, and virtual reality devices like Oculus Rift.

Fig. 1.2 | Logical units of a computer. (Part 2 of 5.)

Logical unit	Description
Memory unit	This rapid-access, relatively low-capacity “warehouse” section retains information that has been entered through the input unit, making it immediately available for processing when needed. The memory unit also retains processed information until it can be placed on output devices by the output unit. Information in the memory unit is <i>volatile</i> —it’s typically lost when the computer’s power is turned off. The memory unit is often called either memory , primary memory or RAM (Random Access Memory). Main memories on desktop and notebook computers contain as much as 128 GB of RAM, though 2 to 16 GB is most common. GB stands for gigabytes; a gigabyte is approximately one billion bytes. A byte is eight bits. A bit is either a 0 or a 1.
Arithmetic and logic unit (ALU)	This “manufacturing” section performs <i>calculations</i> , such as addition, subtraction, multiplication and division. It also contains the <i>decision</i> mechanisms that allow the computer, for example, to compare two items from the memory unit to determine whether they’re equal. In today’s systems, the ALU is implemented as part of the next logical unit, the CPU.

Fig. 1.2 | Logical units of a computer. (Part 3 of 5.)

Logical unit	Description
Central processing unit (CPU)	This “administrative” section coordinates and supervises the operation of the other sections. The CPU tells the input unit when information should be read into the memory unit, tells the ALU when information from the memory unit should be used in calculations and tells the output unit when to send information from the memory unit to certain output devices. Many of today’s computers have multiple CPUs and, hence, can perform many operations simultaneously. A multi-core processor implements multiple processors on a single integrated-circuit chip—a <i>dual-core processor</i> has two CPUs, a <i>quad-core processor</i> has four and an <i>octa-core processor</i> has eight. Today’s desktop computers have processors that can execute billions of instructions per second. To take full advantage of multi-core architecture you need to write multithreaded applications, which we introduce in Section 24.3.

Fig. 1.2 | Logical units of a computer. (Part 4 of 5.)

Logical unit	Description
Secondary storage unit	This is the long-term, high-capacity “warehousing” section. Programs or data not actively being used by the other units normally are placed on secondary storage devices (e.g., your <i>hard drive</i>) until they’re again needed, possibly hours, days, months or even years later. Information on secondary storage devices is <i>persistent</i> —it’s preserved even when the computer’s power is turned off. Secondary storage information takes much longer to access than information in primary memory, but its cost per unit is much less. Examples of secondary storage devices include hard drives, DVD drives and USB flash drives, some of which can hold over 2 TB (TB stands for terabytes; a terabyte is approximately one trillion bytes). Typical hard drives on desktop and notebook computers hold up to 2 TB, and some desktop hard drives can hold up to 6 TB.

Fig. 1.2 | Logical units of a computer. (Part 5 of 5.)

-Data Hierarchy:

Data items processed by computers form a data hierarchy that becomes larger and more complex in structure as we progress from **bits** to **characters** to **fields**, and so on.



Fig. 1.3 | Data hierarchy.

Level	Description
Records	Several related fields can be used to compose a record. In a payroll system, for example, the record for an employee might consist of the following fields (possible types for these fields are shown in parentheses): • Employee identification number (a whole number) • Name (a string of characters) • Address (a string of characters) • Hourly pay rate (a number with a decimal point) • Year-to-date earnings (a number with a decimal point) • Amount of taxes withheld (a number with a decimal point) Thus, a record is a group of related fields. In the preceding example, all the fields belong to the same employee. A company might have many employees and a payroll record for each one.
Files	A file is a group of related records. [<i>Note:</i> More generally, a file contains arbitrary data in arbitrary formats. In some operating systems, a file is viewed simply as a <i>sequence of bytes</i>—any organization of the bytes in a file, such as organizing the data into records, is a view created by the application programmer.] It's not unusual for an organization to have many files, some containing billions, or even trillions, of characters of information.
Database	A database is an electronic collection of data that's organized for easy access and manipulation. The most popular database model is the relational database in which data is stored in simple <i>tables</i>. A table includes <i>records</i> and <i>fields</i>. For example, a table of students might include first name, last name, major, year, student ID number and grade point average. The data for each student is a record, and the individual pieces of information in each record are the fields. You can search, sort and manipulate the data based on its relationship to multiple tables or databases. For example, a university might use data from the student database in combination with databases of courses, on-campus housing, meal plans, etc.

Fig. 1.4 | Levels of the data hierarchy. (Part 2 of 2.)

Unit	Bytes	Which is approximately
1 kilobyte (KB)	1024 bytes	10^3 (1024) bytes exactly
1 megabyte (MB)	1024 kilobytes	10^6 (1,000,000)bytes
1 gigabyte (GB)	1024 megabytes	10^9 (1,000,000,000) bytes
1 terabyte (TB)	1024 gigabytes	10^{12} (1,000,000,000,000) bytes
1 petabyte (PB)	1024 terabytes	10^{15} (1,000,000,000,000,000) bytes
1 exabyte (EB)	1024 petabytes	10^{18} (1,000,000,000,000,000,000) bytes
1 zettabyte (ZB)	1024 exabytes	10^{21} (1,000,000,000,000,000,000,000) bytes

Fig. 1.4 | Byte measurements.



Introduction to C++ Programming, Input/Output and Operators

Chapter 2 of C++ How to Program, 10/e

```
1 // Fig. 2.1: fig02_01.cpp
2 // Text-printing program.
3 #include <iostream> // enables program to output data to the screen
4
5 // function main begins program execution
6 int main() {
7     std::cout << "Welcome to C++!\n"; // display message
8
9     return 0; // indicate that program ended successfully
10 }
```

Welcome to C++!

Fig. 2.1 | Text-printing program.



THANK YOU